

EXPERIMENT NO. : 3 (ASSIGNMENT)

DATE: 27/08/2026

1. AIM

To design and verify a dynamic D flip-flop using True Single-Phase Clock (TSPC) logic in Cadence Virtuoso and to calculate the setup time, clock-to-Q delay, and maximum operating frequency for 1–1 and 0–0 overlapping conditions.

2. TOOL

Cadence Virtuoso

3. THEORY

A True Single-Phase Clock (TSPC) D flip-flop is a dynamic flip-flop that uses a single-phase clock signal for its operation. It consists of cascaded dynamic CMOS stages that alternately perform precharge and evaluation according to the clock level.

During the precharge phase, the internal dynamic nodes are charged or discharged to their required logic levels. During the evaluation phase, the input data is evaluated and propagated through the dynamic stages to produce the corresponding output Q. Since only one clock signal is required, TSPC logic reduces clocking complexity and can provide high-speed operation.

The setup time (t_{setup}) is the minimum time for which the input D must remain stable before the active clock edge for correct data capture. The clock-to-Q delay (t_{C-Q}) is the time interval between the active clock edge and the corresponding transition at the output Q.

For reliable operation, the minimum clock period is given by $T_{min} = t_{setup} + t_{C-Q}$. Hence, the maximum operating frequency is $f_{max} = 1/T_{min}$. These timing parameters are determined for both 1–1 and 0–0 overlapping conditions using transient simulation in Cadence Virtuoso.

The TSPC D flip-flop is simulated using 180 nm CMOS technology, and the obtained waveforms are used to calculate the setup time, clock-to-Q delay, minimum clock period, and maximum operating frequency.

4. TABLES

Table 4.1: Design Parameters

Parameter	Value	Unit
Technology	180	nm
V_{dd}	1.8	V
PMOS Width	400	nm
NMOS Width	400	nm

Table 4.2: Result comparison (pre layout vs post layout) of Transmission Gate based dff

parameter	pre layout	post layout	unit	error
t_{setup}	108	171	ns	+58.3333%
t_{C-Q}	337.9	718.5	ns	+112.6369%
t_{min}	445.9	889.5	ps	+99.4842%
f_{max}	2.2427	1.1242	GHz	-49.8729%

5. CALCULATION FOR TRANSMISSION GATE BASED D FLIP FLOP

$$t_{setup(pre)} = (27.0005 - 26.8925) * 1000 = 108ps$$

$$t_{setup(post)} = (27.0005 - 26.8295) * 1000 = 171ps$$

$$t_{C-Q(pre)} = (27.3384 - 27.0005) * 1000 = 337.9ps$$

$$t_{C-Q(post)} = (27.7190 - 27.0005) * 1000 = 718.5ps$$

$$T_{min(pre)} = t_{setup(pre)} + t_{C-Q(pre)} = 108 + 337.9 = 445.9ps$$

$$T_{min(post)} = t_{setup(post)} + t_{C-Q(post)} = 171 + 718.5 = 889.5ps$$

$$f_{max(pre)} = 1/T_{min(pre)} * 1000 = 2.243GHz$$

$$f_{max(post)} = 1/T_{min(post)} * 1000 = 1.124GHz$$

6. CIRCUIT DIAGRAMS & WAVEFORMS

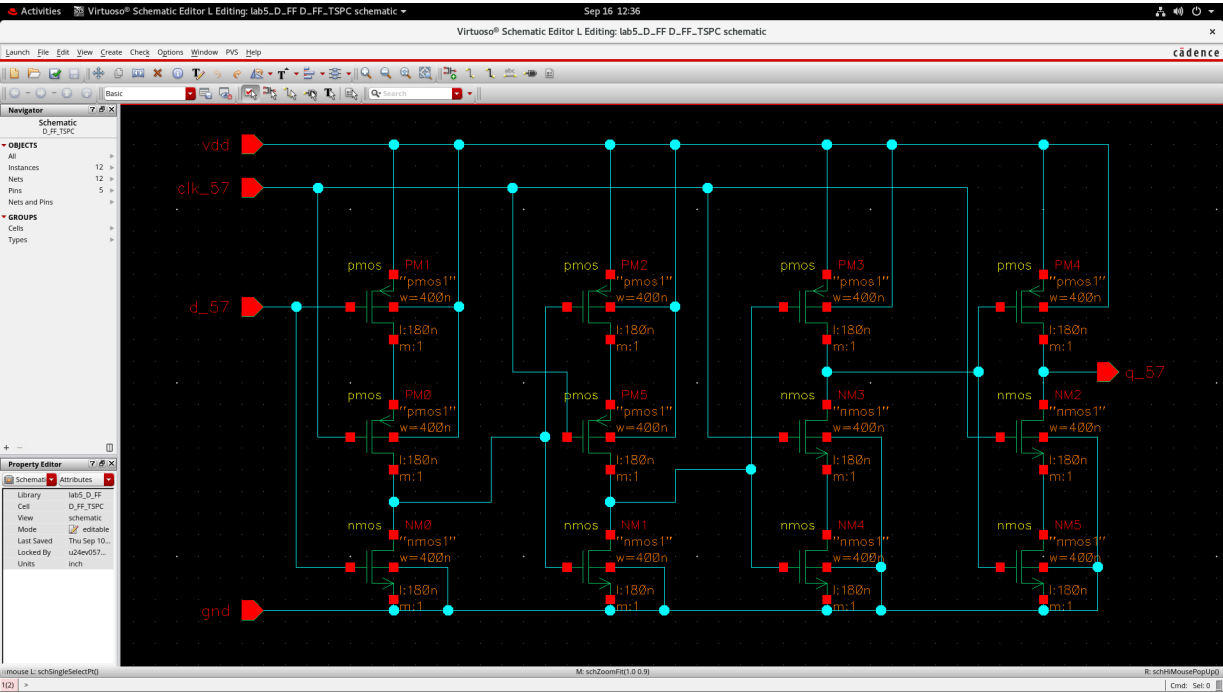


Figure 6.1: Tspc Schematic

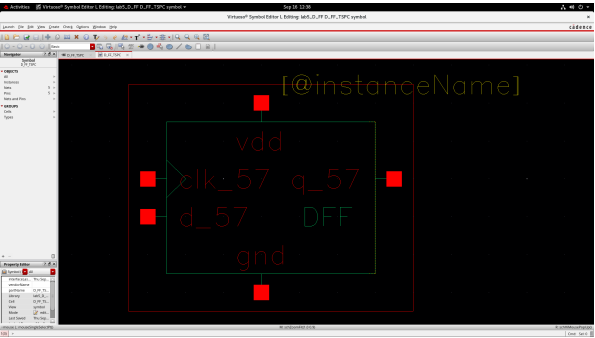


Figure 6.2: Tspc Symbol

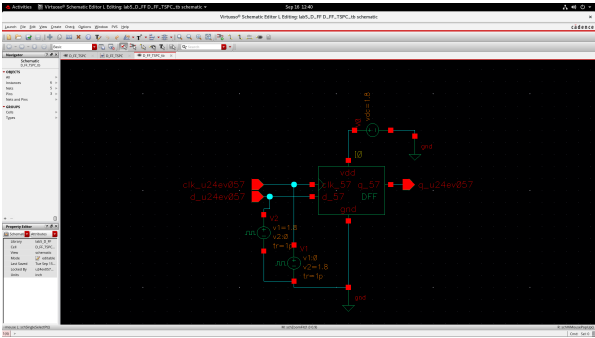


Figure 6.3: Tspc Tb

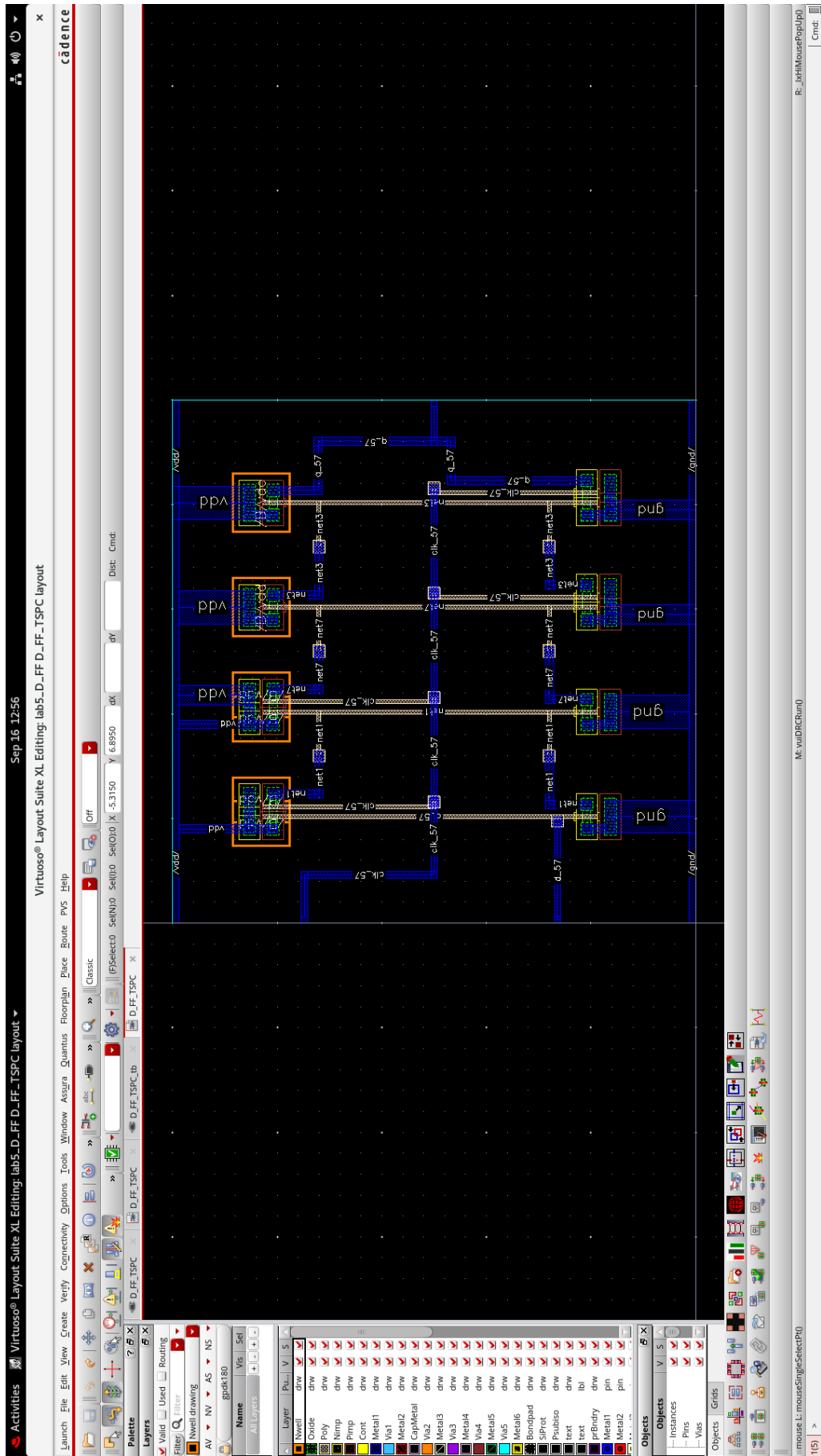


Figure 6.4: Tspc Layout

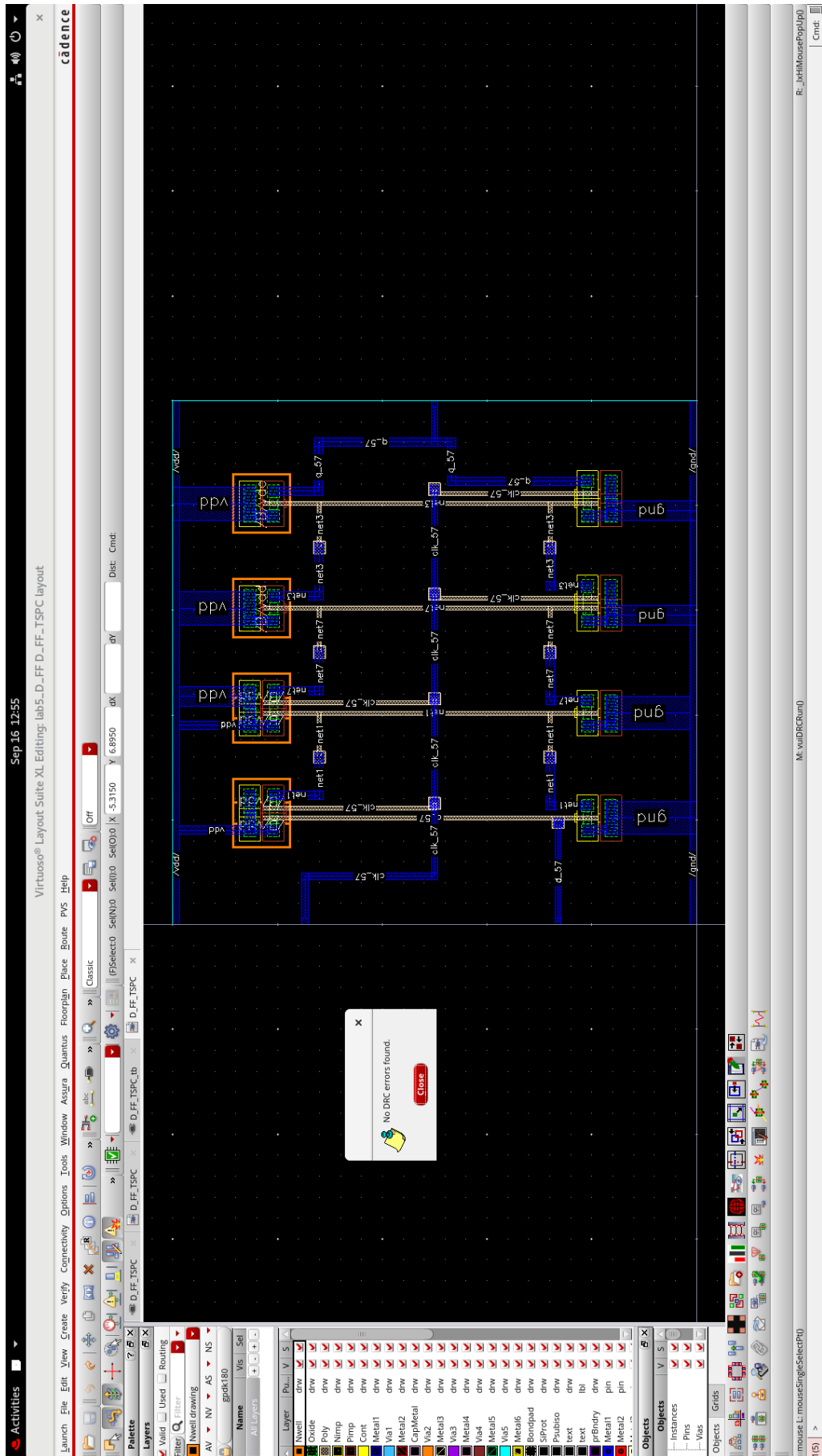


Figure 6.5: Tspc Drc

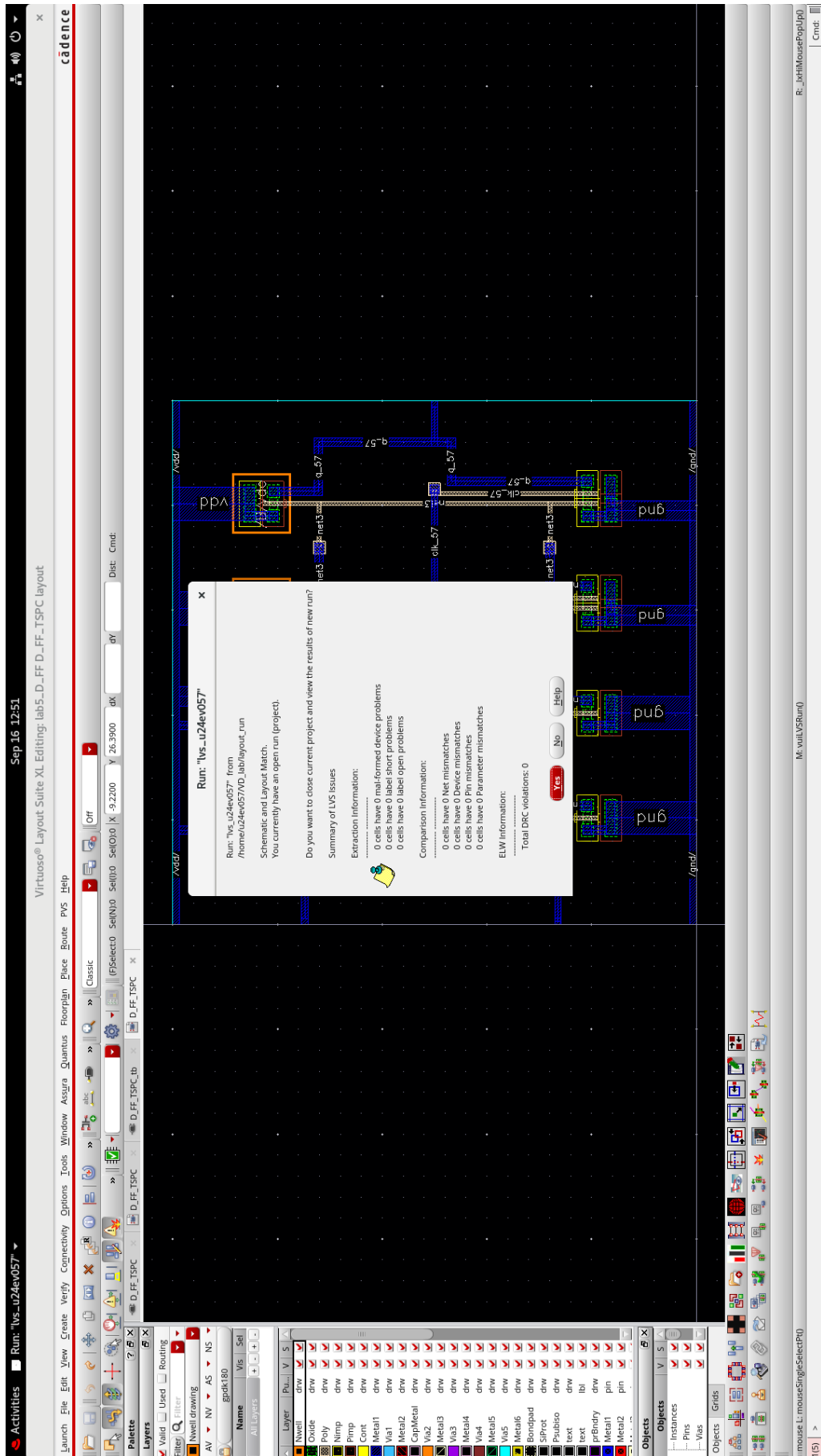


Figure 6.6: Tspc Lvs

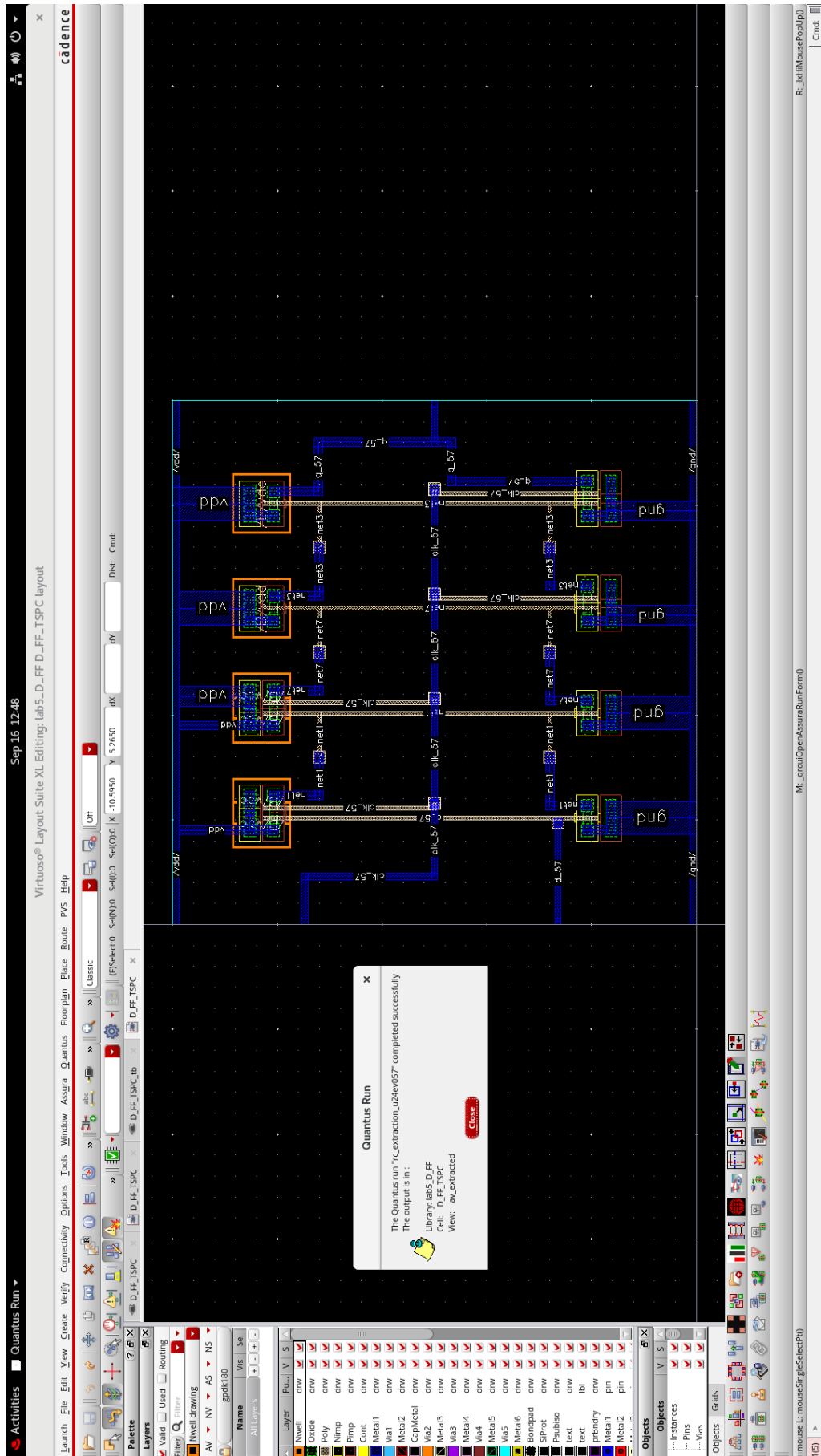


Figure 6.7: Tspc Quantus

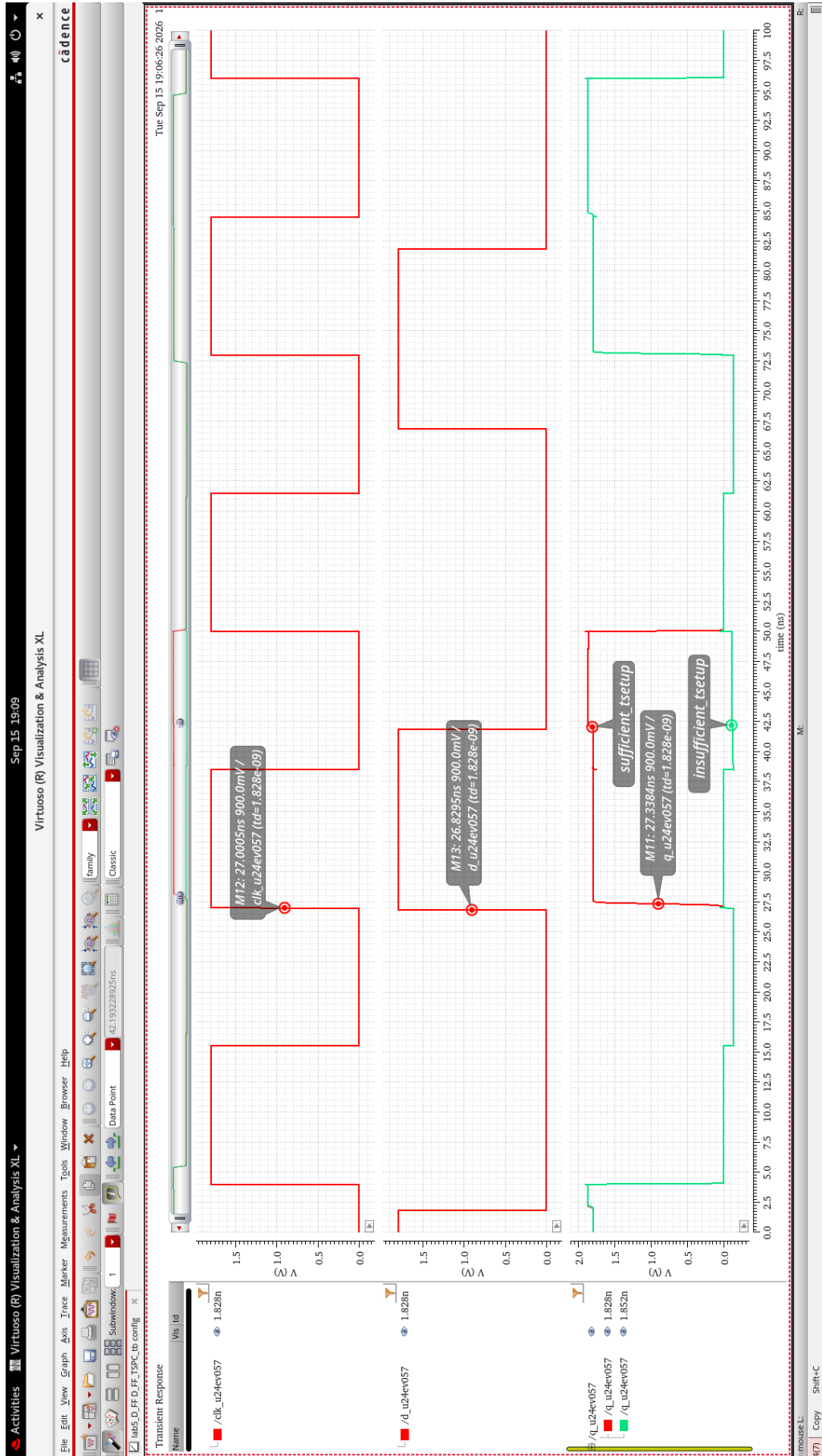


Figure 6.8: Tspc Tran Pre Layout

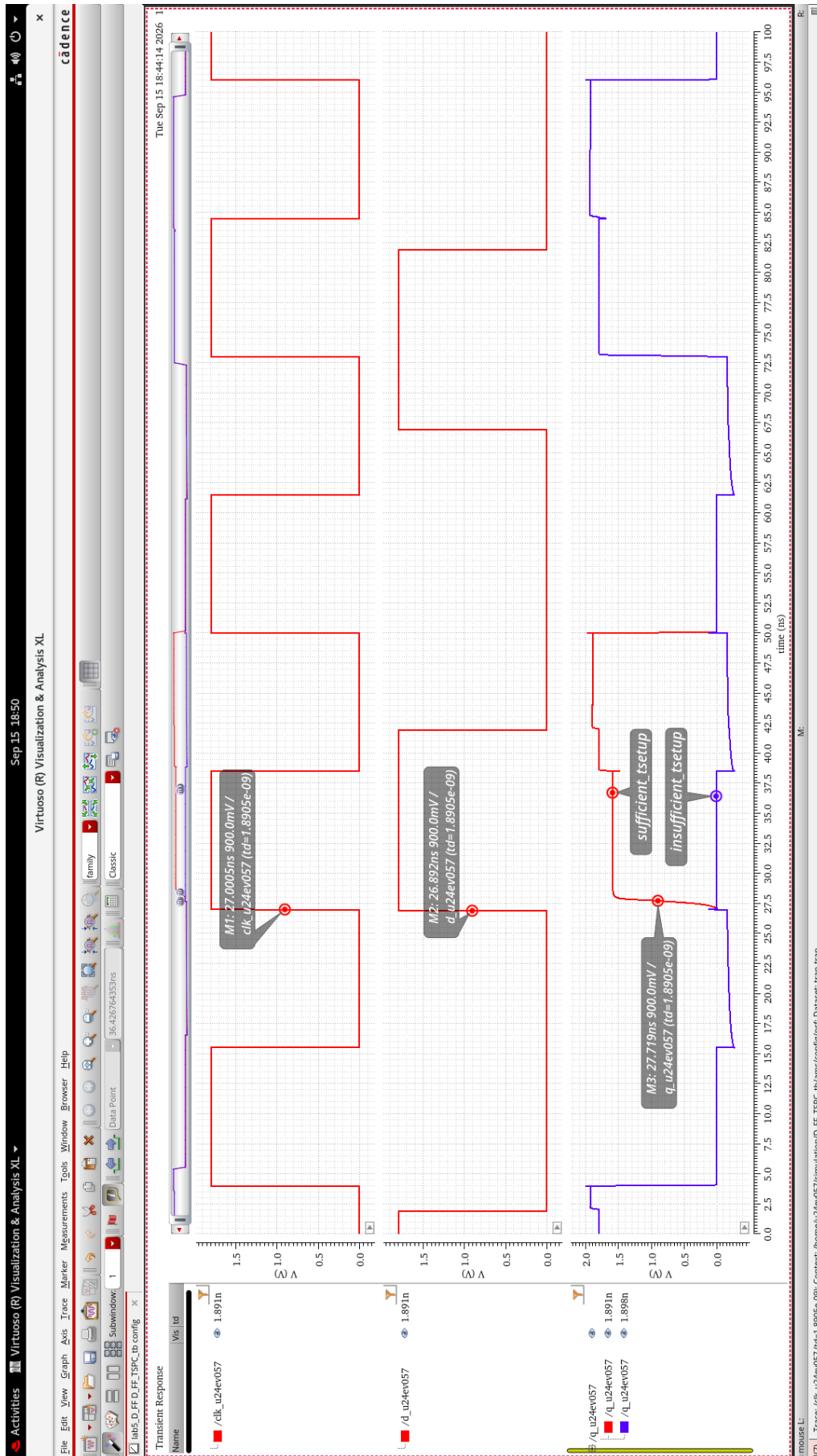


Figure 6.9: Tspc Tran Post Layout

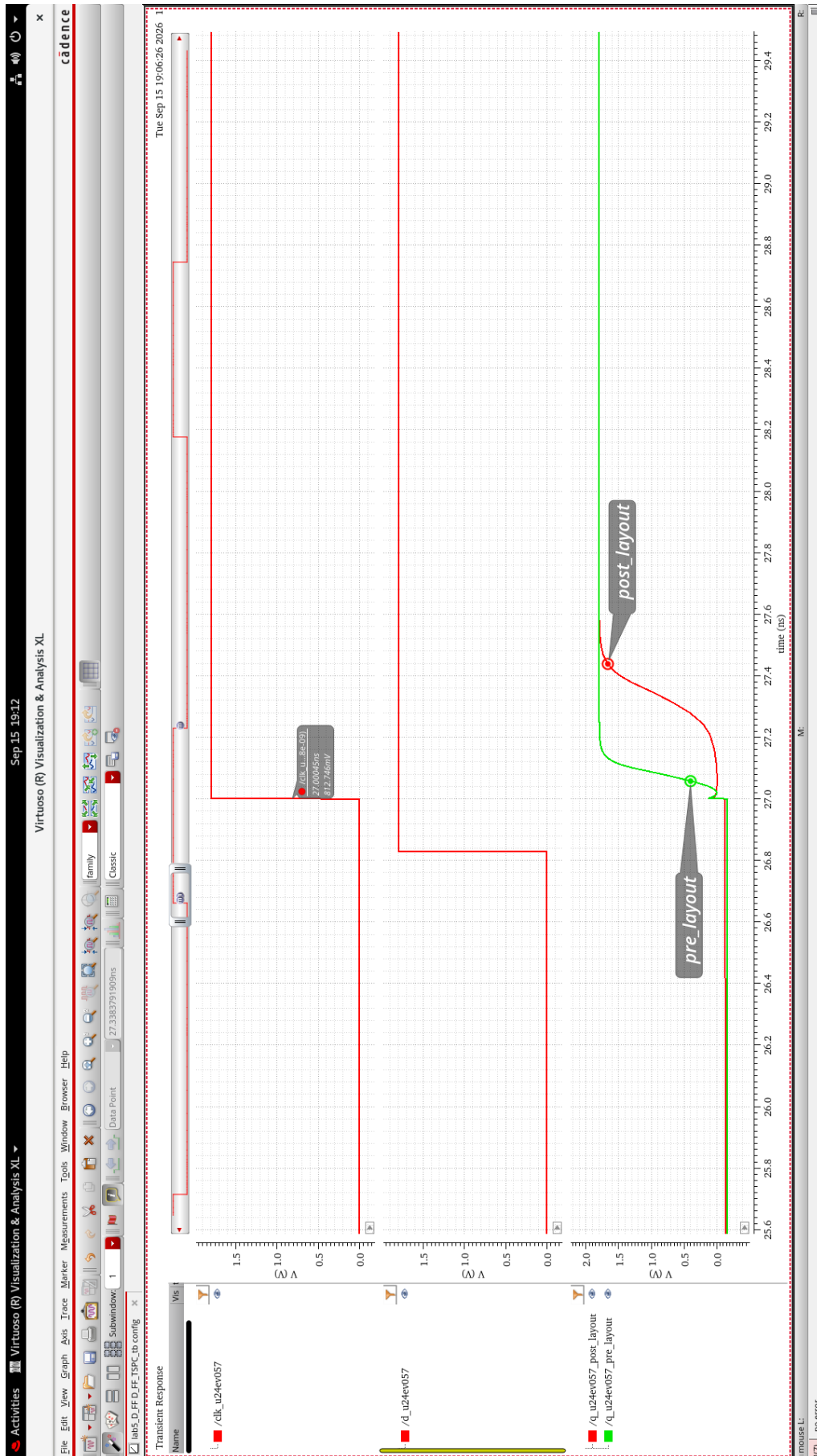


Figure 6.10: Tspc Tran Pre Vs Post

7. RESULT

The TSPC dynamic D flip-flop was successfully designed and verified. The setup time increased from 108 ps to 171 ps and the clock-to-Q delay increased from 337.9 ps to 718.5 ps after layout, reducing the maximum operating frequency from 2.243 GHz to 1.124 GHz.